

Maharashtra 2020**Grade 12****Chemistry****Series: J-346****SET - E****Time allowed: 3 hours****Maximum Marks: 70****General Instructions:**

The question paper is divided into four sections.

- (1) Section A: Q. No 1 concludes **Ten** multiple choice type of questions carrying One mark each.
Q. No 2 contains **Eight** very short answer type of questions carrying One mark each.
- (2) Section B: Q. No. 3 to Q. No. 14 are **Twelve** short answer type of questions carrying Two marks each (Attempt any Eight)
- (3) Section C: Q. No 15 to Q. 26 are **Twelve** short answer type of questions carrying Three marks each (Attempt any Eight)
- (4) Section D: Q. No 27 to Q. Bo. 31 are **Five** long answer type of questions carrying Four marks each (Attempt any Three)
- (5) Use of log table is allowed. Use of calculator is not allowed.
- (6) Figures to the right indicate full marks.
- (7) For each MCQ. Correct answer must be written along with its alphabet. e.g. (a)/ (b)/ (c)/ (d) etc.
- (8) Physical constant:

$$\text{Avogadro's Number} = N_A = 6.022 \times 10^{23}$$

SECTION - A

Q.1. Select and write correct answer of the following questions:

[10]

1. Identify synthetic polymer amongst the following:

- (a) Linear
- (b) Jute
- (c) Silk
- (d) Terylene

2. Which among the following hydrides is NOT a reducing agent?

- (a) H_2O
- (b) H_2S
- (c) H_2Te
- (d) H_2Se

3. During oxidation of ferrous sulphate using mixture of dil. H_2SO_4 and potassium dichromate, oxidation state of chromium changes from _____.

- (a) +6 to +2
- (b) +6 to +3
- (c) +6 to +1
- (d) +6 to +4

4. Identify complex ion in which effective atomic number of the central metal ion is 35.

(Given At. Number of Co = 27, Fe = 26, Zn = 30)

- (a) $[\text{Zn}(\text{NH}_3)_6]^{2+}$
- (b) $[\text{Zn}(\text{CN})_6]^{4-}$
- (c) $[\text{Zn}(\text{CN})_6]^{3-}$
- (d) $[\text{Co}(\text{NH}_3)_6]^{3+}$

5. Conversion of methyl chloride into methyl fluoride is known as _____.

- (a) Finkelstein reaction (b) Swarts reaction
(c) Williamson's synthesis (d) Wurtz reaction

6. The number of moles of methyl iodide required to prepare tetramethyl ammonium iodide from 1 mole of methyl amine leave:

- (a) 1 (b) 2
(c) 3 (d) 4

7. Name the reagent which on reaction with glucose confirms the presence of five hydroxyl groups in glucose:

- (a) Hydroxyl amine
(b) Bromine water
(c) Dilute nitric acid
(d) Acetic anhydride

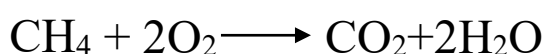
8. The number of moles of methyl iodide required to prepare tetramethyl ammonium iodide from 1 mole of methyl amine leave:

- (a) Codeine (b) Equanil
(c) Penicillin (d) Valium

9. The number of atoms per unit cell body centered cube is:

- (a) 1 (b) 2
(c) 4 (d) 6

10. Calculate the work done during the reaction represented by the following thermochemical equation at 300 K:



(g) (g) (g) (l)

($R = 8.314 \text{ JK}^{-1}\text{mol}^{-1}$)

(a) +4.988 KJ (b) -4.988 KJ

(c) -49.88 KJ (d) +49.88 KJ

Q.2. Answer the following questions:

[8]

(i) What is the concentration of dissolved oxygen at 50°C under pressure of one atmosphere if partial pressure of oxygen at 50°C is 0.14 atm.

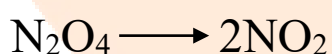
(ii) Write structural formula of the alcohol that results when acetaldehyde is reacted with CH_3MgBr in the presence of dry ether and the product is hydrolysed.

(iii) Write the chemical reaction for preparation of acetic anhydride using acetic acid.

(iv) Write the chemical reaction involved in the formation of ethyl amine using acetaldoxime.

(v) What is electrometallurgy?

(vi) For the reaction:

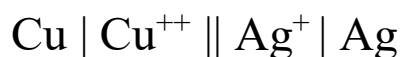


(g) (g)

($\Delta H^{\circ} = +57.24 \text{ KJ}$, $\Delta S^{\circ} = +175.8 \text{ JK}^{-1}$)

At what temperature the reaction will be spontaneous?

(vii) The standard e.m.f. of the following cell is 0.463 V



(s) (1M) (1M) (s)

If the standard potential of Ag electrode is 0.800 V, what is the standard potential of Cu Electrode?

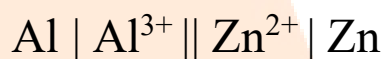
(viii) Write the mathematical relation between half life of zero order reaction and its rate constant.

SECTION – B

Attempt any EIGHT of the following questions: [16]

Q. 3. State and explain Hess's law of constant heat summation.

Q. 4. Write the cell reaction and calculate E^0 cell of the following electrochemical cell:



(s) (aq.) (aq.) (s)
(1M) (1M)

$$E^0_{\text{Al}} = -1.66\text{V}$$

$$E^0_{\text{Zn}} = -0.76\text{V}$$

Q. 5. Distinguish between order and molecularity of a reaction.

Q. 6. Write two uses of each of the following:

(a) Helium

(b) Neon

Q. 7. Write the name and chemical formula of one ore of zinc.

Define : Quaternary ammonium salt.

Q. 8. What is the action of acidified potassium dichromate on the following:

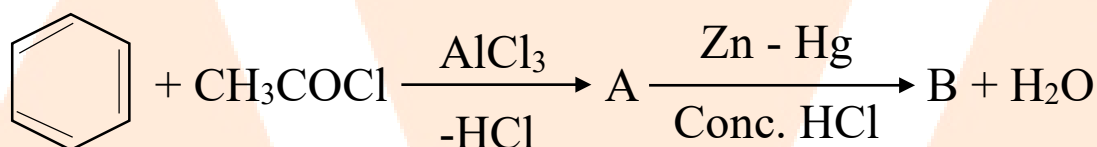
(a) KI

(b) H₂S

Q. 9. Define optical activity. How many optical isomers are possible for glucose?

Q. 10. Explain continuous etherification process for the preparation of diethyl ether.

Q. 11. Identify 'A' and 'B' in the following reaction:



Q. 12. Write Haworth projection formula of
 α - D - (+) - glucopyranose

Define hormones.

Q. 13. Classify the following solids into different types:

- | | |
|-------------|--------------------|
| (a) Silver | (b) P ₄ |
| (c) Diamond | (d) NaCl |

Q. 14. Define:

- (a) Molality
(b) Osmotic pressure

SECTION – C

Attempt any EIGHT of the following questions:

[24]

Q. 15. Define flux.

Write a note on leaching process.

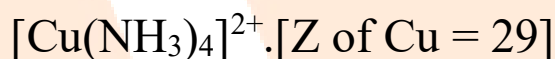
Q. 16. Draw the structure of sulphurous acid.

Explain why nitrogen does not form pent halides.

Q. 17. Write the general electronic configuration of lanthanoids.

Why are most of the compounds of transition metals coloured?

Q. 18. Calculate the effective atomic number (e.a.n.) of copper in



Explain ionisation isomerism in coordination compounds with a suitable example.

Q. 19. Write the chemical reactions of chlorobenzene with respect to:

- (a) Sulphonation
- (b) Acetylation
- (c) Nitration

Q. 20. How is ethanol prepared from the following compounds?

- (a) Ethanol
- (b) Ethene
- (c) Bromoethane

Q. 21. How primary, secondary and tertiary alcohols are distinguished using HNO_2 ?

Q. 22. What are monosaccharides? Explain denaturation of proteins.

Q. 23. Define non-biodegradable polymer.

Write the preparation of terylene.

Q. 24. What are soaps? How are soaps prepared?

Define antiseptic.

Q. 25. Unit cell of a metal has edge length of 288 pm and density of 7.86 g cm^{-3} . Determine the type of crystal lattice.

[Atomic mass of metal = 56 g mol^{-1}]

Q. 26. Define instantaneous rate of reaction.

Explain pseudo first order reaction with suitable example.

SECTION – D

Attempt any THREE of the following questions:

[12]

Q. 27. Define the terms:

(a) Electrochemical series

(b) Corrosion

Write two applications of electrochemical series.

Q. 28. Explain interhalogen compounds.

How is oxygen prepared from the following compounds?

(a) KClO_4

(b) PbO_2

Q. 29. Explain the mechanism of aldol addition reaction.

Mention two uses of carboxylic acids.

Q. 30. Derive the mathematical expression between molar mass of a non-volatile solute and elevation of boiling point.

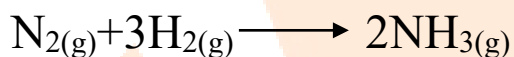
State and explain van't Hoff-Avogadro's law.

Q. 31. Define:

(a) Reversible process.

(b) Standard enthalpy of combustion

Calculate the enthalpy change for the reaction:



The bond enthalpies are:

Bond	N = N	H - H	N - H
$\Delta H^\circ \text{KJmol}^{-1}$	946	435	389